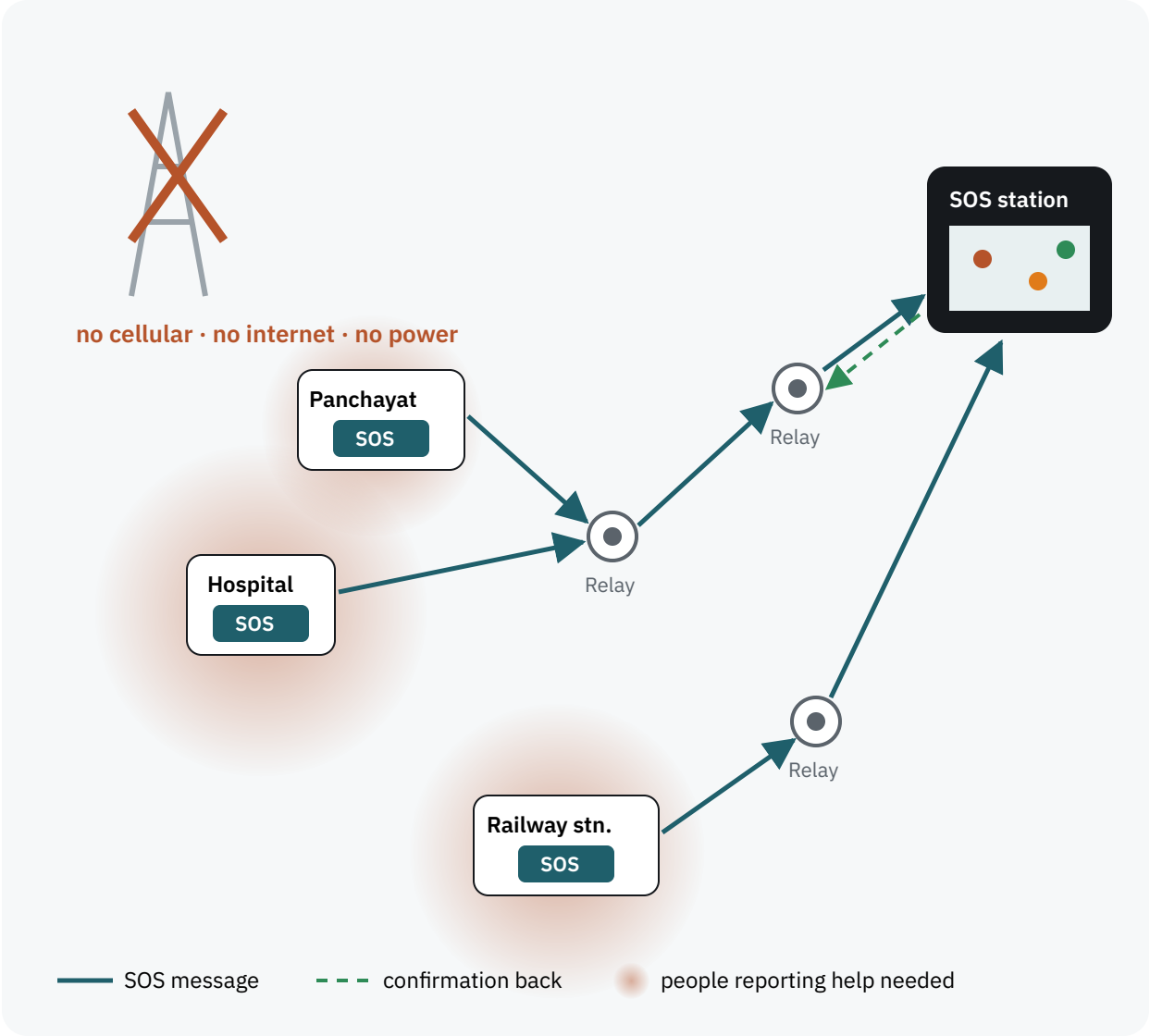


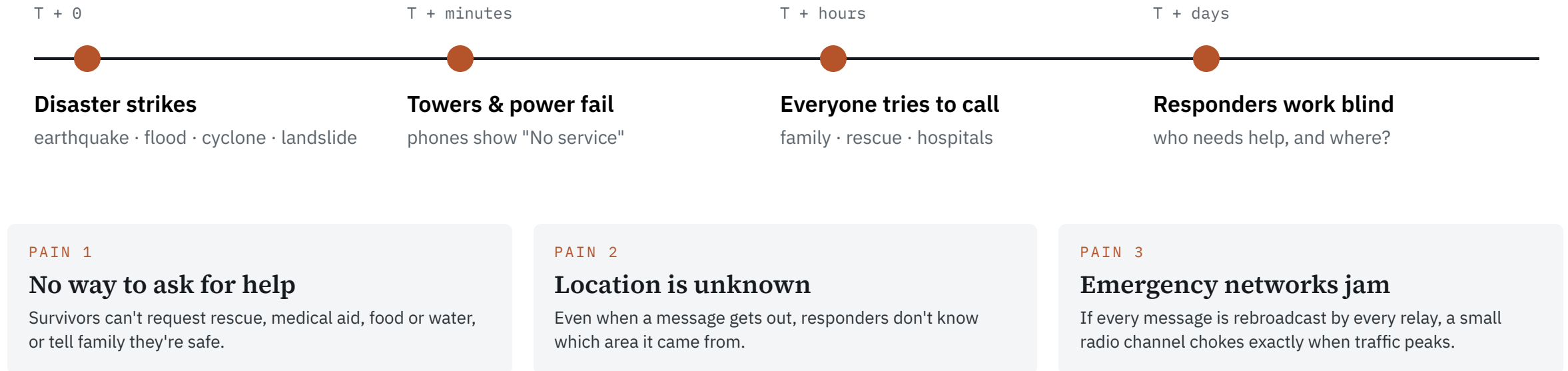
VOID-NAV

When the towers fall, every hospital, panchayat and railway station becomes a lifeline.

Public **SOS nodes** + a controlled **LoRa relay network** let survivors call for help and reach family with no cellular network, no internet and no app. Every message is **prioritised**, kept **congestion-free**, and its sender sees an **orange** or **green** light confirming what happened to it.

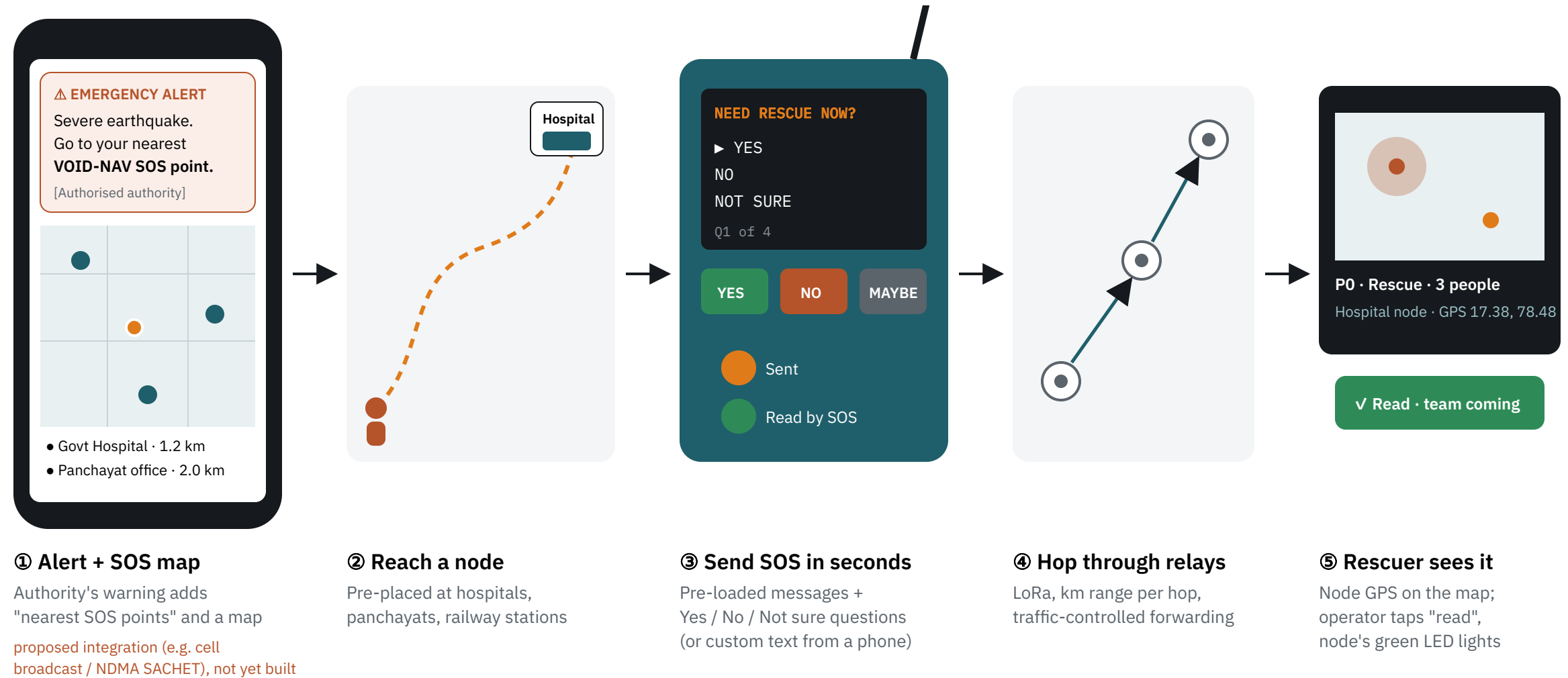


After a disaster, the first thing everyone wants is to talk. That's exactly when the network disappears.

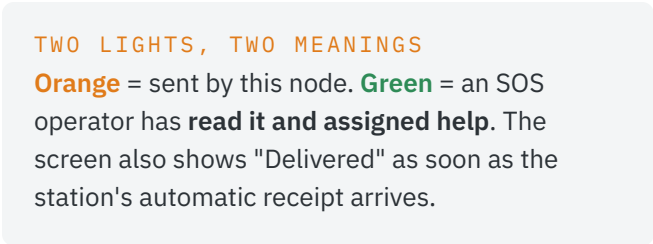
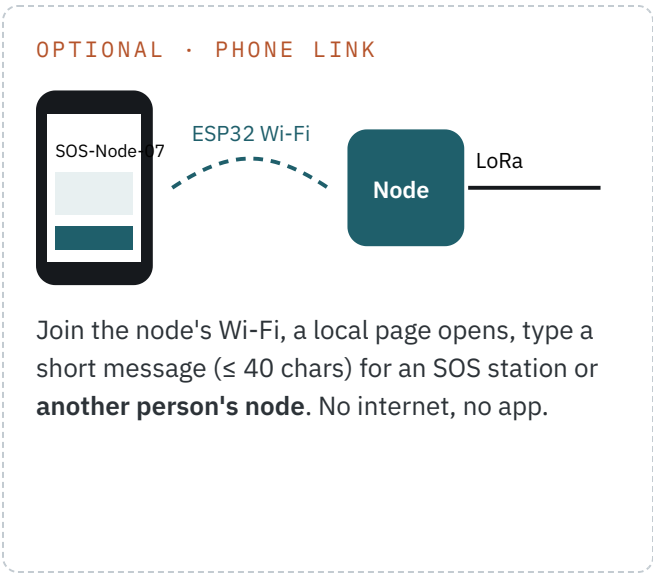
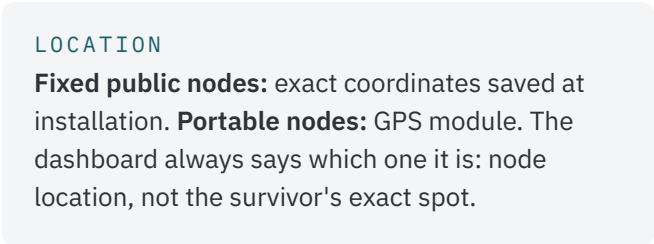
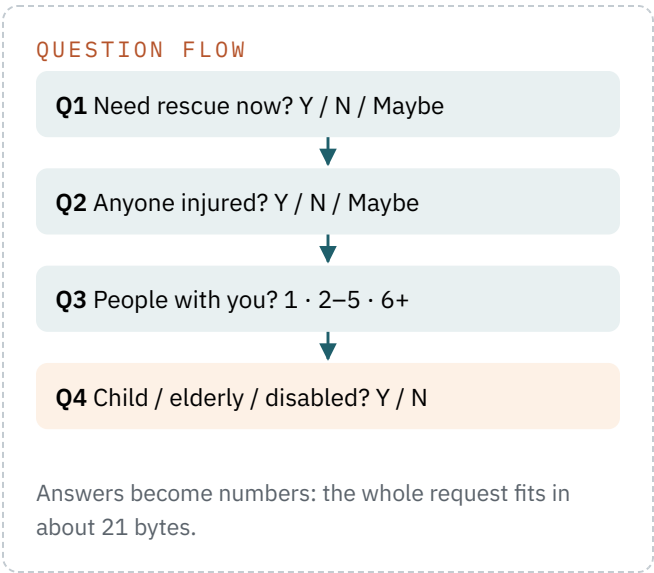
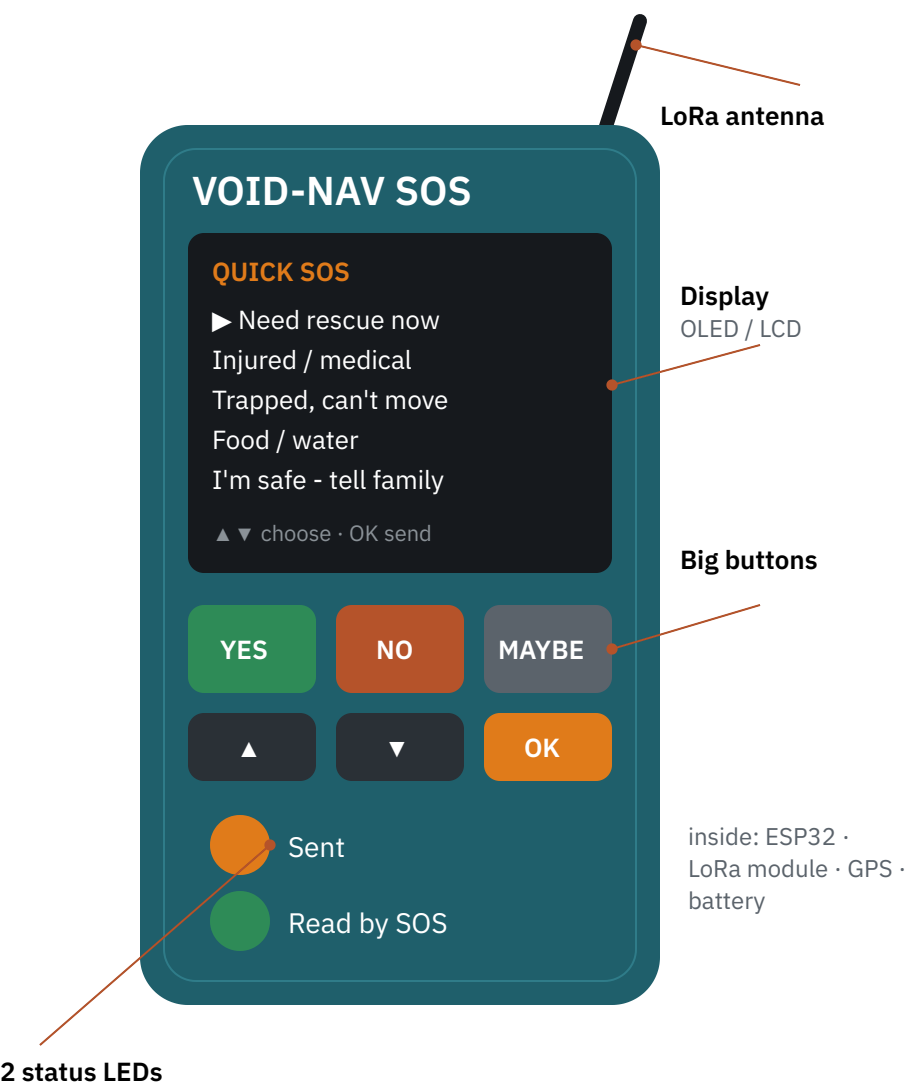


We need a communication path that works with **zero infrastructure**, is **simple for a panicked person**, tells the sender their message **arrived**, and **stays fast** even when hundreds of people use it at once.

From alert to rescue

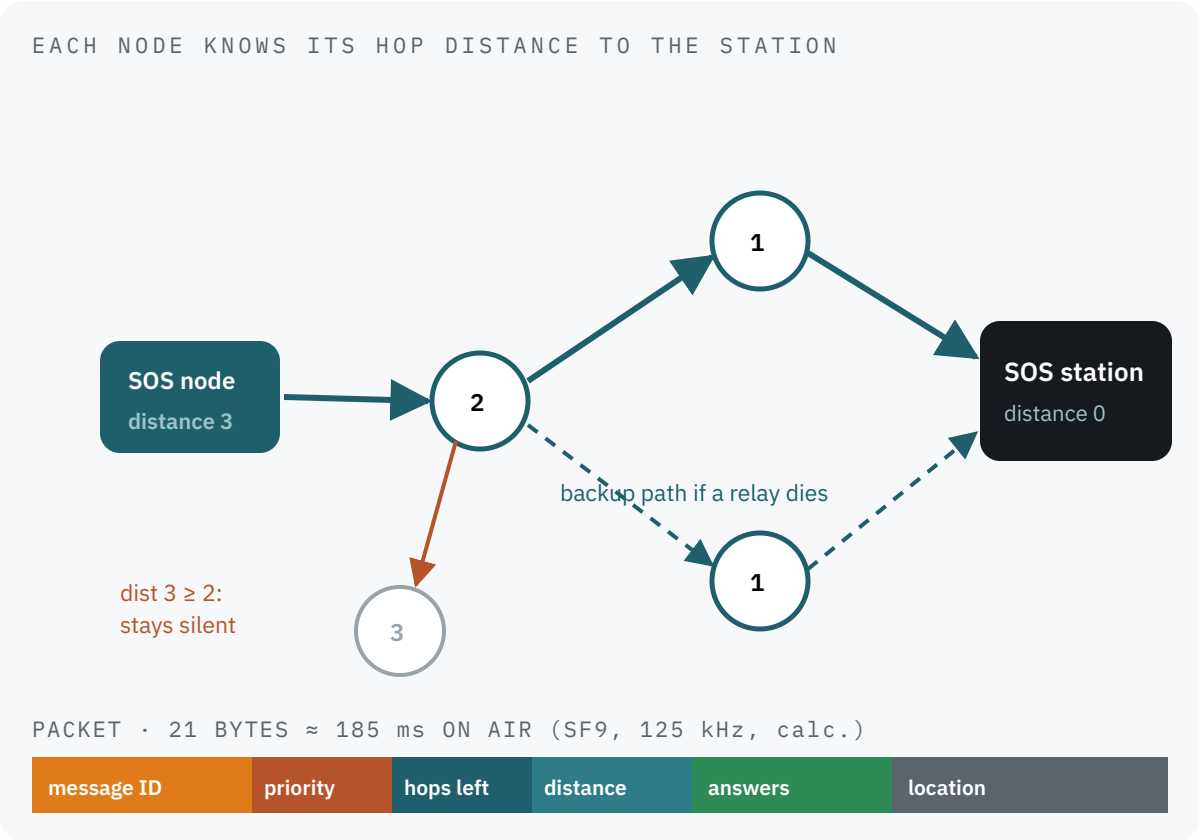


A device anyone can use in a panic



Fast for urgent cases, never jammed

LoRa doesn't create a mesh by itself. We wrote the forwarding rules; these six stop the channel from choking.



- 1 · FORGET REPEATS

Every message has a unique ID. Relays remember recent IDs and drop copies.
- 2 · HOP LIMIT + EXPIRY

Max 4 hops and a lifetime (10–30 min), so nothing circulates forever.
- 3 · CLOSER-ONLY FORWARDING

A relay forwards only if it is **nearer the station** than the sender. No blind flooding.
- 4 · PRIORITY LANES

P0 rescue/medical → P1 vulnerable → P2 food/water → P3 "I'm safe". Waiting messages move up a lane so none starve.
- 5 · LISTEN, THEN TALK

Check the channel, wait a short random time (shorter for P0), and cancel if another relay already forwarded it.
- 6 · CAPPED RETRIES

No receipt? Retry with growing gaps, max 5 for P0 and 2 for P3, reusing the same ID.
- ≈ 0.3 s**

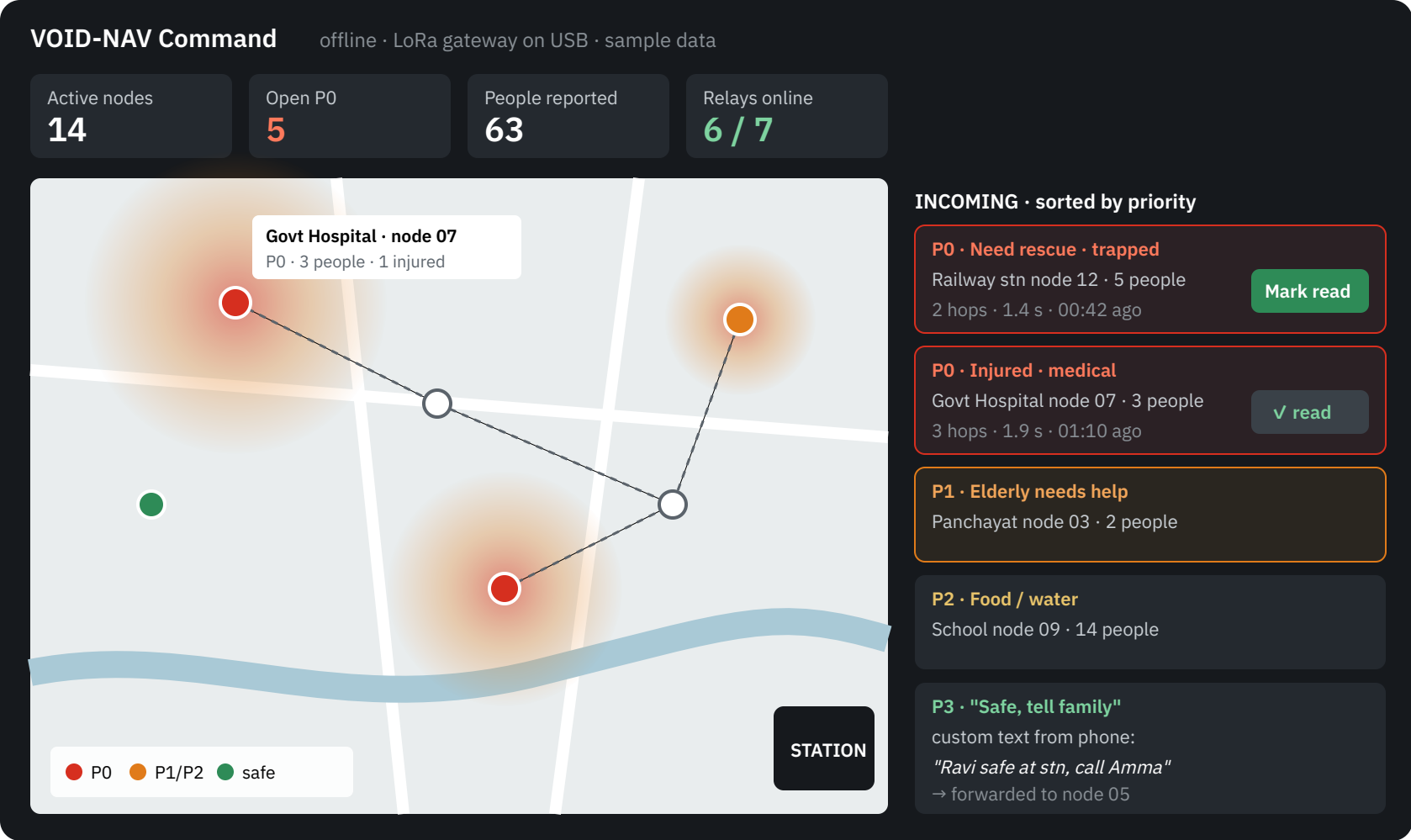
per hop for P0 (estimate)
- ≈ 1.6 s**

3 hops there + receipt back, idle channel (est.)
- 1 channel**

shared: more relays ≠ more capacity

Estimates from Semtech's airtime formula; to be measured on our prototype. Collisions are reduced, not eliminated.

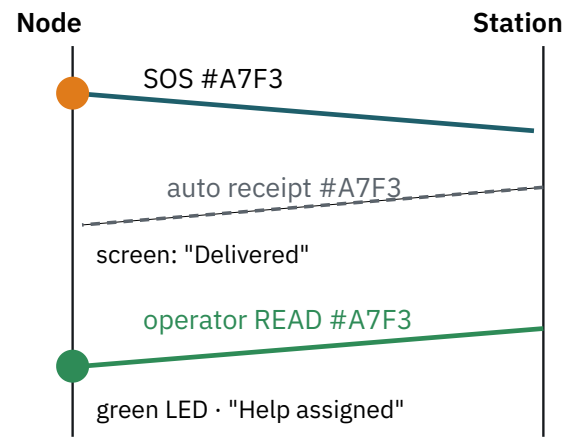
Rescuers see who needs help, where, and how urgent



WHAT THE DASHBOARD SHOWS

- **Live map** with every SOS node, relay and link health
- **Survivor heatmap**: brightness = people reported per area (from SOS answers), not tracking
- **Priority queue**: P0 always on top
- **Location label**: fixed-node coordinates or node GPS
- **"Mark read"** sends the green-light message back

CONFIRMATION ROUND TRIP



Built for the messy reality of disasters

SURVIVOR'S PHONE IS DEAD OR LOST

Use the node directly

Buttons + screen only. No phone, no app, no literacy in English needed: menus can be in the local language.

PHONE WORKS, NO NETWORK

Phone → node Wi-Fi

Type a custom message on the node's local page; it travels over LoRa like any SOS.

FAMILY FAR AWAY

Node-to-node messages

Address a message to another SOS point; it routes toward that node only, not flooded everywhere.

RESCUERS HAVE PHONES

Field view on Wi-Fi

The station laptop serves the dashboard on its own local Wi-Fi; teams nearby open it on their phones.

RESCUERS' PHONES ARE OFF

Station is enough

The control room reads the dashboard and briefs teams by radio; a portable node can go with each team.

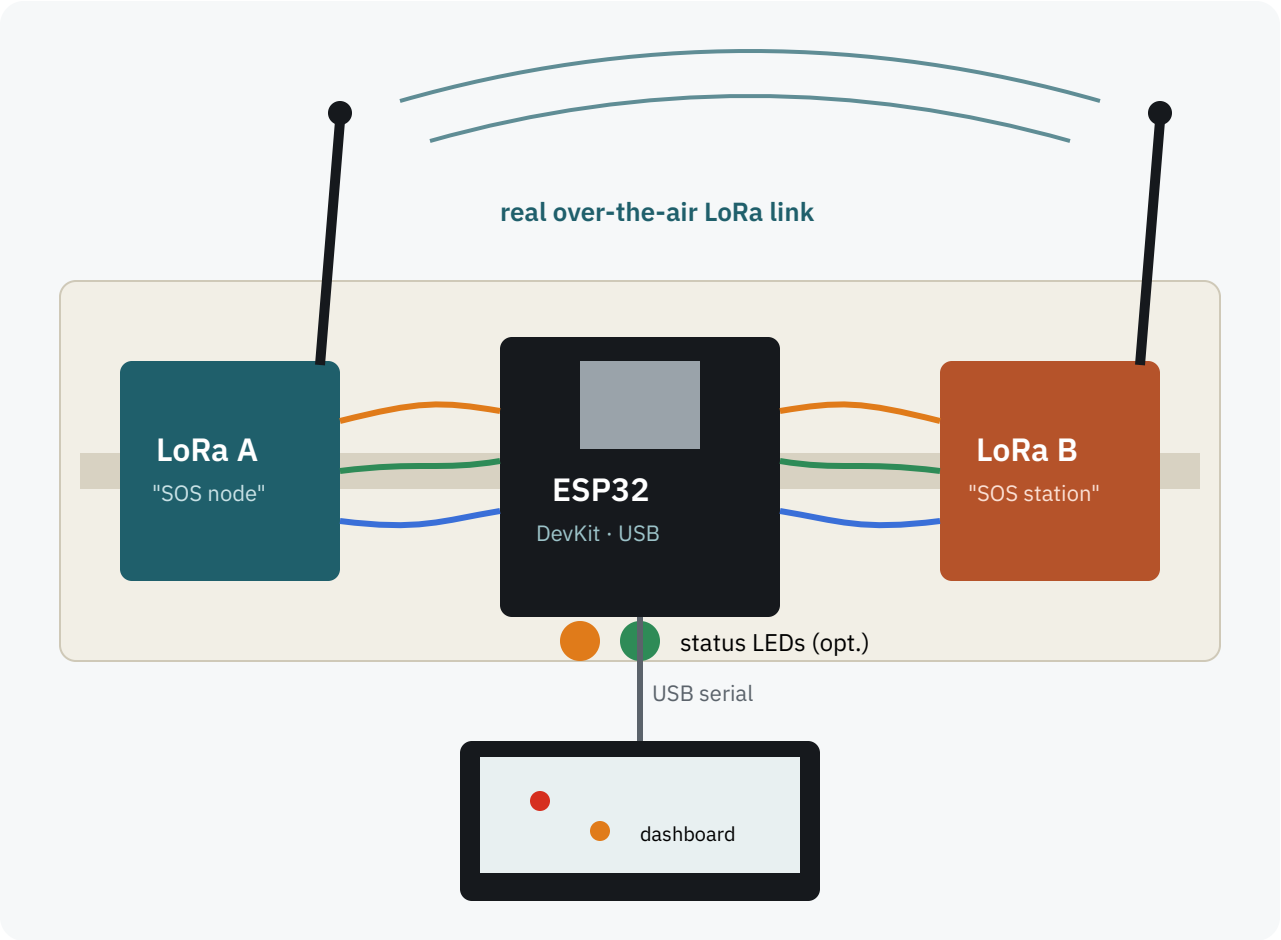
SOME CELLULAR COMES BACK

Optional bridge

The station can forward summaries to the district control room when internet returns. LoRa stays the backbone.

DISASTER	WHERE NODES & RELAYS GO	SPECIAL CARE
Earthquake	open grounds, hospitals, panchayat offices; relays on poles away from damaged buildings	shock-proof cases, pre-charged batteries
Flood	rooftops, elevated water tanks, railway platforms; relays high up	waterproof enclosures, floating/elevated mounts
Cyclone	cyclone shelters, schools, railway stations; relays installed before landfall	wind-rated low mounts, solar + battery
Landslide	village centres on both sides of a blocked road; relay chain along the valley	portable nodes carried in by rescue teams

Our demo kit: 1 ESP32, 2 LoRa radios, 2 antennas



WIRING · BOTH RADIOS SHARE ONE SPI BUS

SIGNAL	LORA A	LORA B
VCC / GND	3.3 V / GND (never 5 V)	
SCK · MISO · MOSI	GPIO 18 · 19 · 23 (shared)	
NSS (chip select)	GPIO 5	GPIO 4
RESET	GPIO 14	GPIO 27
DIO0 (interrupt)	GPIO 26	GPIO 25

Both modules must be the same frequency band, with antennas fitted before any transmission.

DEMO SCRIPT

1. Serial menu acts as the node screen: pick "Need rescue", answer Y/N/Maybe
2. LoRa A sends the 21-byte packet over the air; **orange** lights
3. LoRa B receives it, decodes it, and the dashboard pin appears
4. Click "Mark read"; B sends it back; A receives it; **green** lights
5. Send the same ID twice: the duplicate is dropped; P0 jumps ahead of P3 in the queue

Honest scope: one ESP32 drives both radios, so this proves the packet, receipt, LED, priority and duplicate logic over a real radio link. A true multi-hop relay chain needs 1 ESP32 + 1 LoRa per node (≥ 3 boards); that's our next build step.

Not for the radio link, but it can plan where relays go

WHERE QUANTUM DOESN'T BELONG

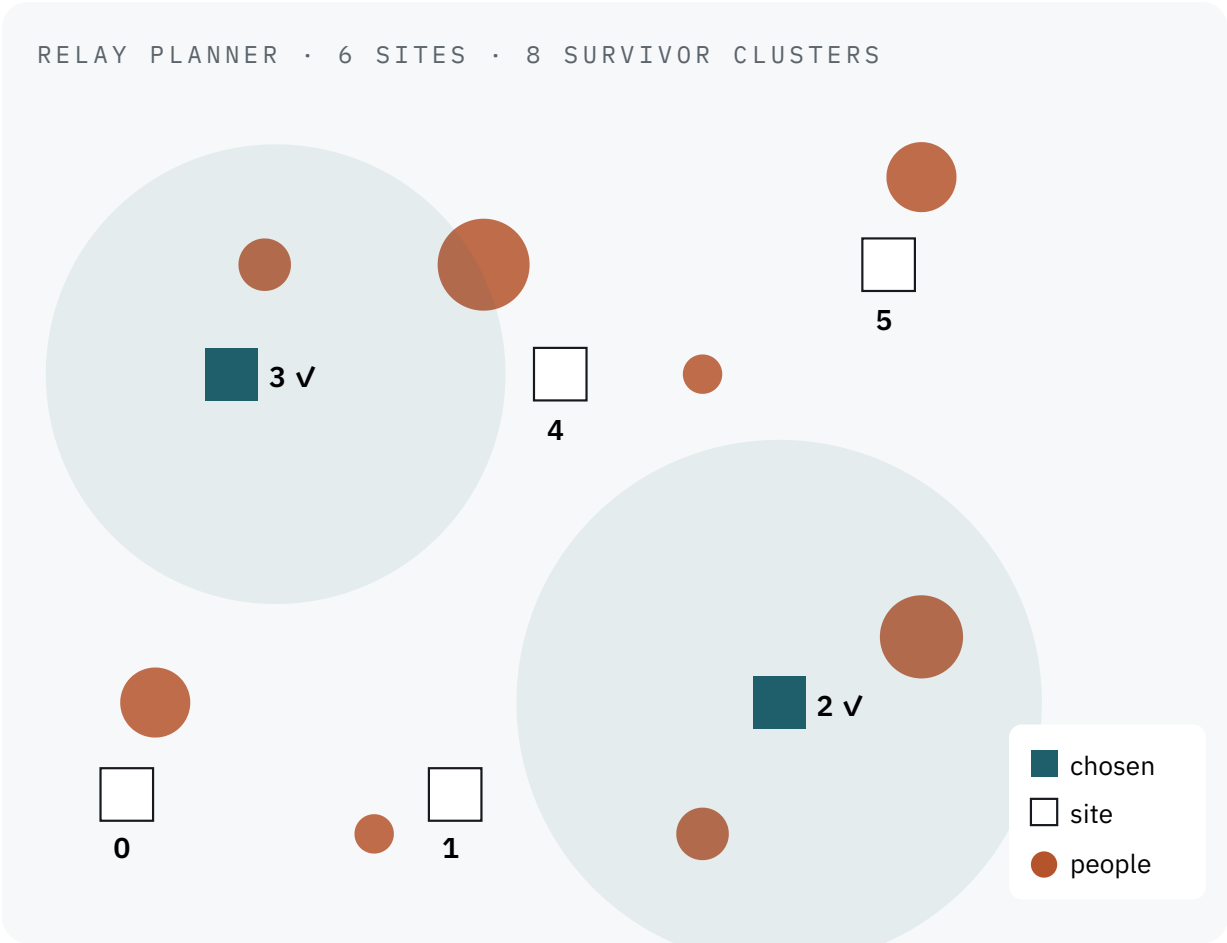
Sending SOS packets, relaying, receipts, LEDs: this all runs on cheap microcontrollers in milliseconds. Adding quantum here would add cost and nothing else. **The core system is fully classical.**

WHERE IT FITS: PLANNING BEFORE A DISASTER

With a limited number of relays, *where* you put them decides how many people stay connected. That's a "choose K of N sites" optimisation that grows explosively with map size. We formulate it as a **QUBO** and solve it with **QAOA on Qiskit**, as an offline planning tool.

OUR QAOA TEST (QISKIT AER)	RESULT
6 candidate sites · place 2 relays · p = 2	6 qubits · COBYLA
QAOA best placement	sites 2 + 3 → 13 of 21 people
Exact (brute force)	same answer ✓
Chance of sampling the best answer	3.7 % vs 1.6 % random

At 6 sites a laptop is just as fast. We show the method and its correctness, not a speed-up. Optional add-on, not core.



A communication lifeline when everything else is down

0

dependence on towers, internet, cloud or apps

≤ 4 presses

to send a complete SOS from a node

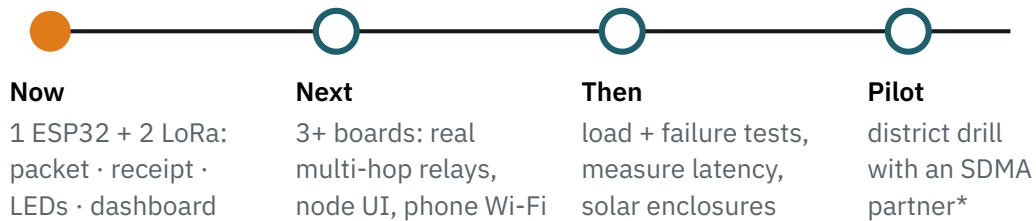
2 lights

so survivors know: sent, then read and help assigned

WHO BENEFITS

Survivors	ask for help and reach family without signal
Rescue teams	a prioritised, located list instead of searching blind
District authorities	live picture of needs: rescue, medical, food, water
Public facilities	hospitals, panchayats, stations become comms points

ROADMAP



*proposed; alert integration and partners are goals, not existing agreements

"When the towers fall, every SOS point becomes a lifeline."

Team VOID
THANK YOU